SDT621 Software design testing C#:

Name	Student Number	Role
Jandre Van Zyl	20230660	Member 1 — Project Manager / GitHub Coordinator
Dewald Van Jaarsveldt	20230258	Member 2 — UI Developer
Tyler Davel	20230614	Member 3 — Logic Developer
Vehan Kuhn	20230275	Member 4 — Tester & Documentation Specialist

Introduction:

This document presents the testing and documentation for the Ticket Application System developed using C# Windows Forms.

The purpose of this document is to demonstrate that the system meets all functional and non-functional requirements. It includes test cases, validation results, screenshots, and GitHub contribution evidence.

The system was tested to ensure accurate ticket calculations, correct application of discount rules, and proper handling of user input.

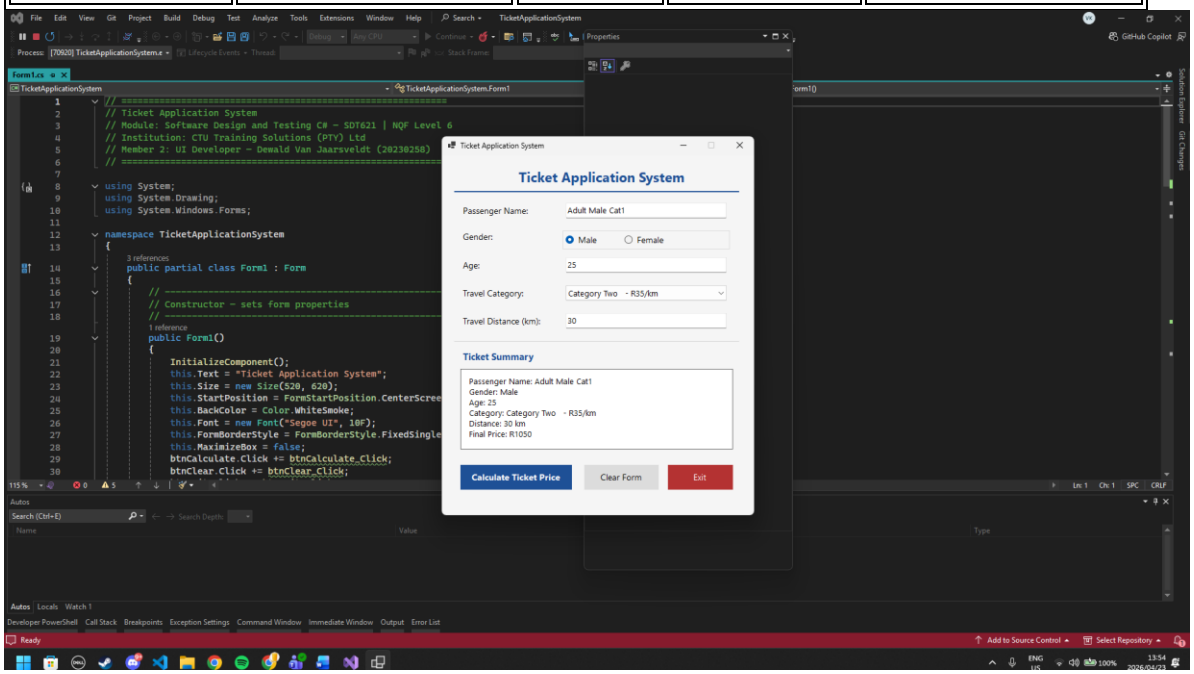
Test plan:

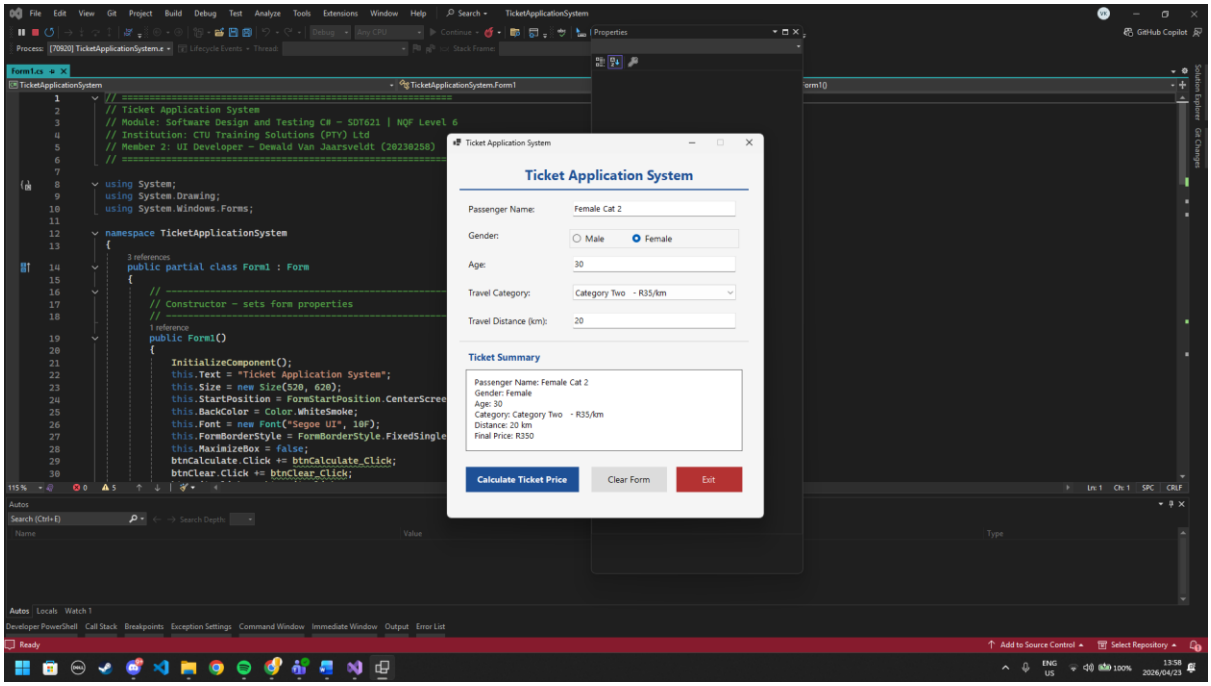
The system was tested to ensure:

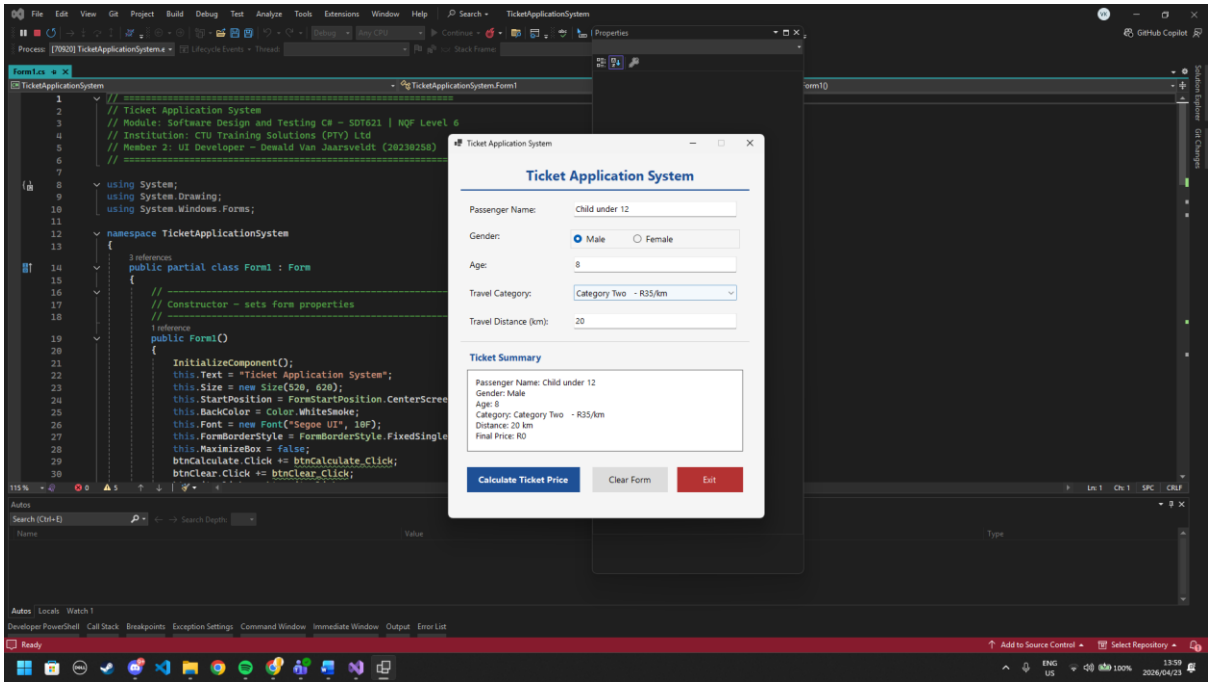
- Correct ticket calculations
- Proper discount application
- Input validation
- Button functionality

Test Cases Table:

Test Case	Input	Expected Result	Actual Result	Status
Adult Male Cat 1	Age 25, 30km, Category Two	R1050	R1050	Functioning correctly
Female Cat 2	Age 30, 20km Category Two	R350	R350	Functioning correctly
Child under 12	Age 8 Male, 20km, Category Two	FREE	FREE	Functioning correctly




// ==========
// Ticket Application System
// Module: Software Design and Testing C# – SDT621 | NQF Level 6
// Institution: CTU Training Solutions (PTY) Ltd
// Member 2: UI Developer – Dewald Van Jaarsveldt (20230258)
// ==========
using System;
using System.Drawing;
using System.Windows.Forms;
namespace TicketApplicationSystem
{
3 references
public partial class Form1 : Form
{
// ----------
// Constructor – sets form properties
// ----------
1 reference
public Form1()
{
InitializeComponent();
this.Text = "Ticket Application System";
this.Size = new Size(520, 620);
this.StartPosition = FormStartPosition.CenterScree
this.BackColor = Color.WhiteSmoke;
this.Font = new Font("Segoe UI", 10F);
this.FormBorderStyle = FormBorderStyle.FixedSingle
this.MaximizeBox = false;
btnCalculate.Click += btnCalculate_Click;
btnClear.Click += btnClear_Click;
Ticket Application System
Ticket Application System
Passenger Name:
Female Cat 2
Gender:
Male
Female
Age:
30
Travel Category:
Category Two - R35/km
Travel Distance (km):
20
Ticket Summary
Passenger Name: Female Cat 2
Gender: Female
Age: 30
Category: Category Two - R35/km
Distance: 20 km
Final Price: R350
Calculate Ticket Price
Clear Form
Exit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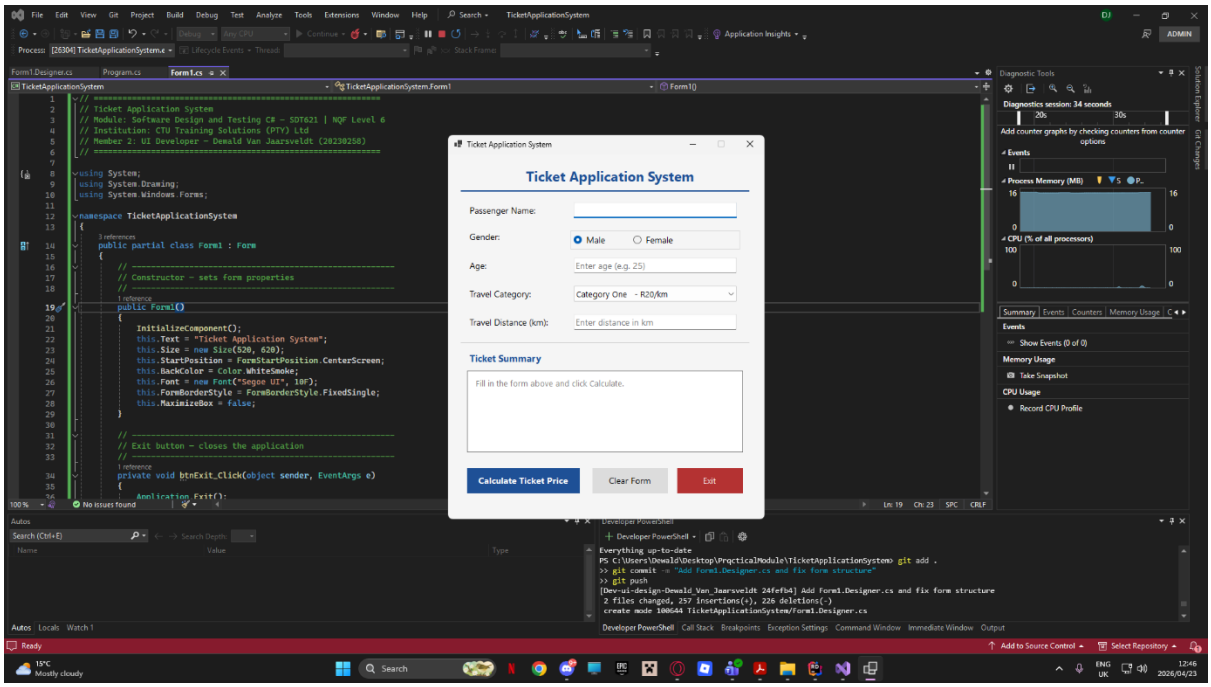
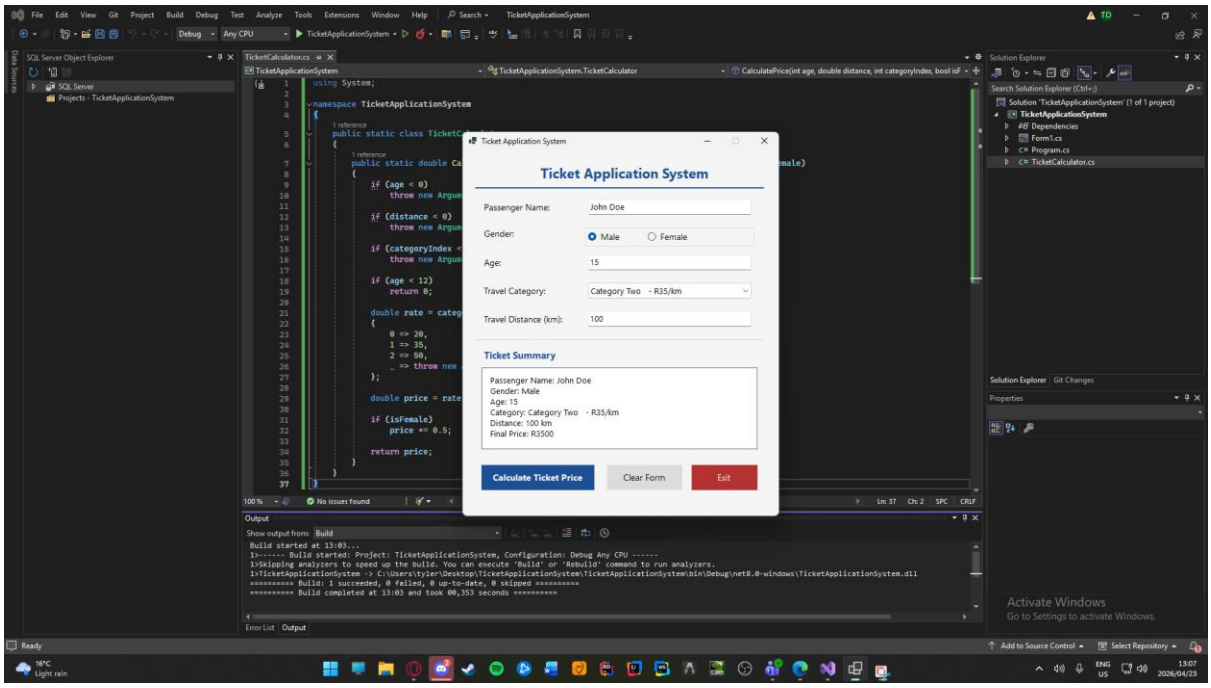
btnClear.Click += btnClear_Click;
Ticket Application System
Ticket Application System
Passenger Name:
Child under 12
Gender:
Male
Female
Age:
8
Travel Category:
Category Two - R35/km
Travel Distance (km):
20
Ticket Summary
Passenger Name: Child under 12
Gender: Male
Age: 8
Category: Category Two - R35/km
Distance: 20 km
Final Price: R0
Calculate Ticket Price
Clear Form
Exit

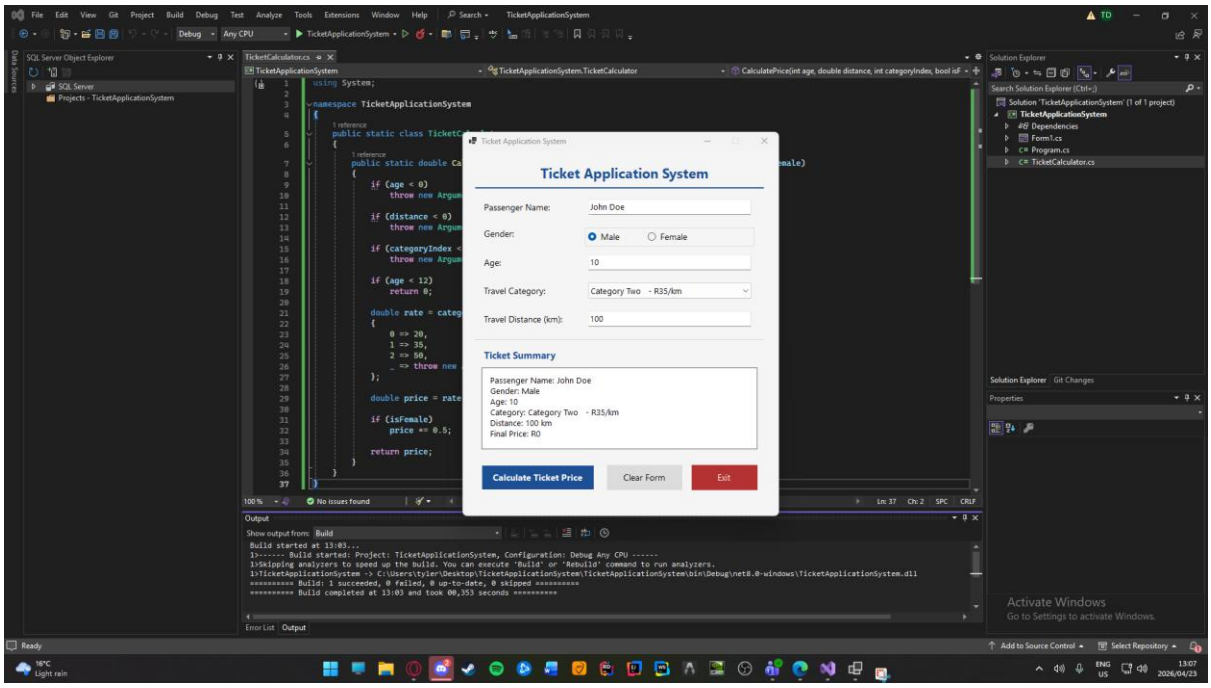
Screenshots:

Screenshot 1: Application UI



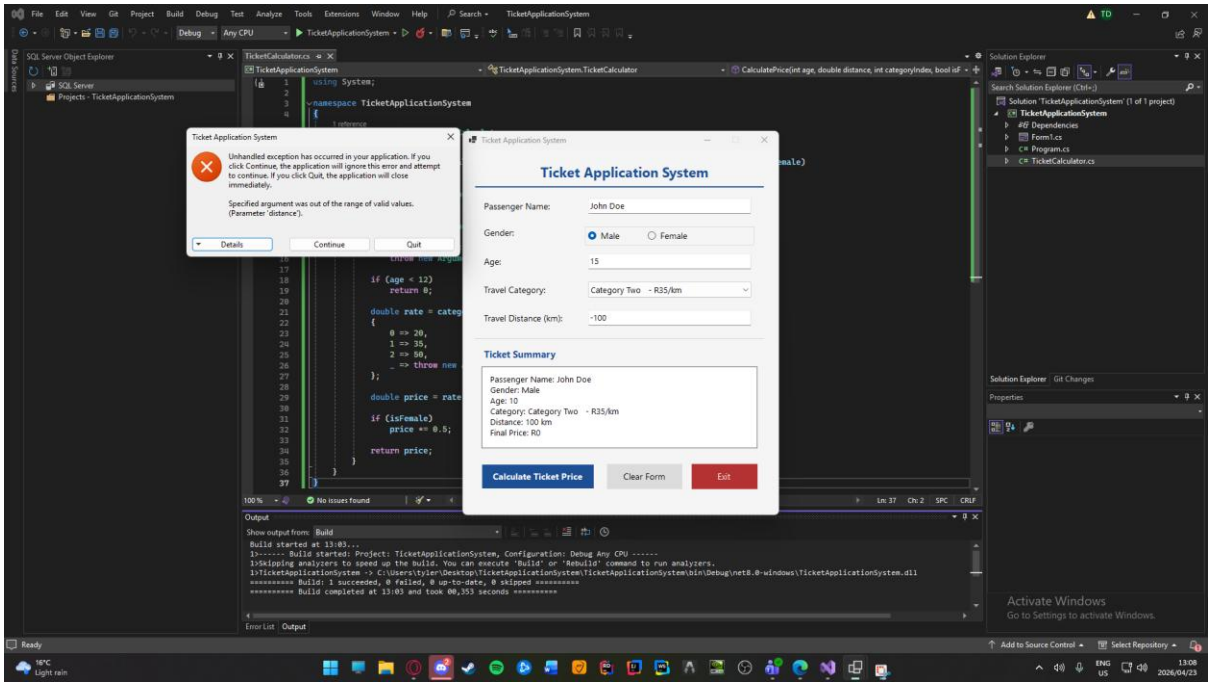
Screenshot 2: Calculation Result



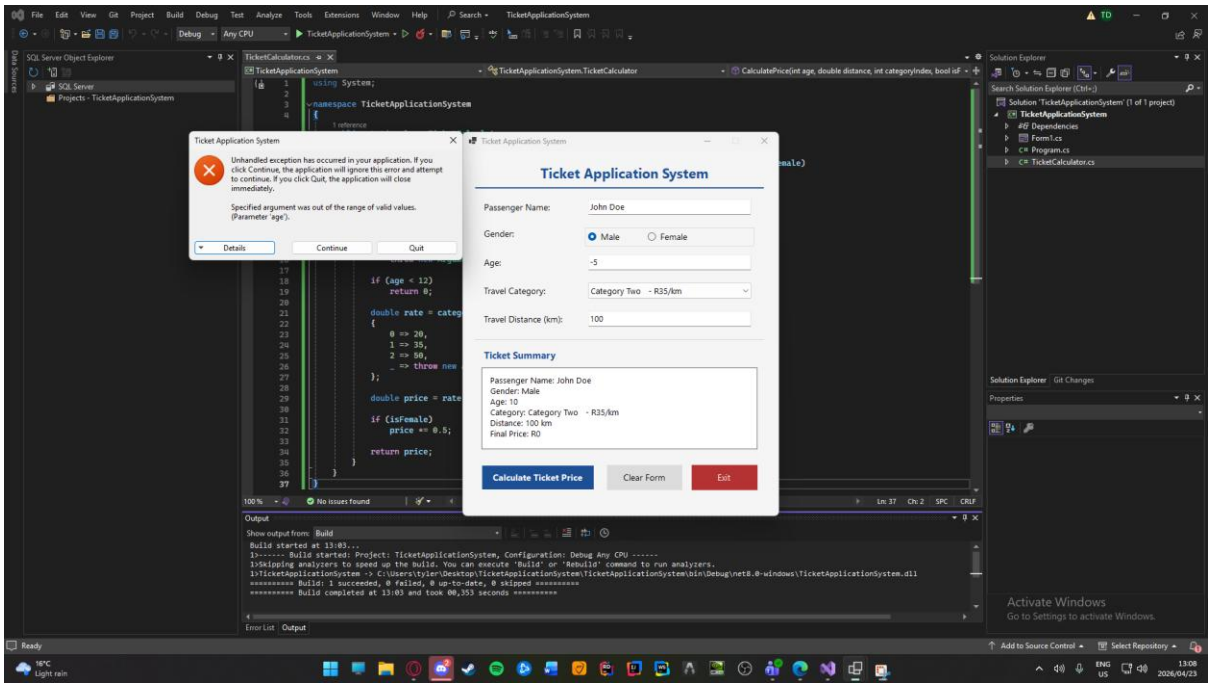


Screenshot 3: Validation Error

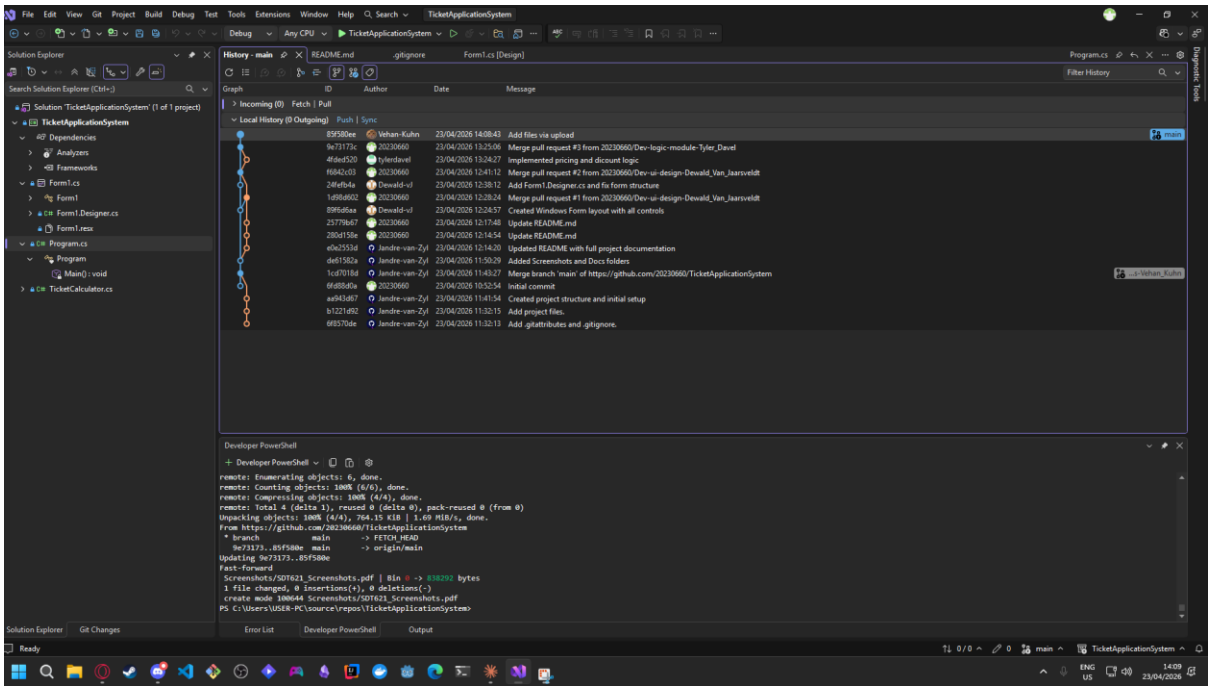
Distance Validation Error:



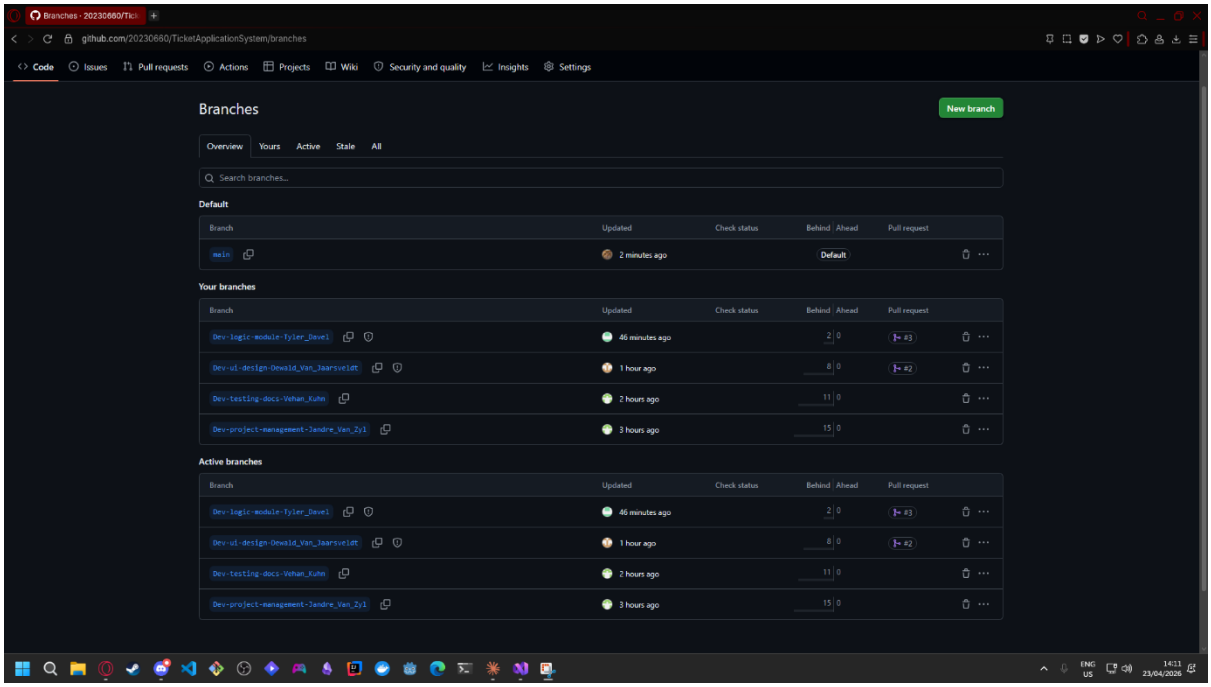
Incorrect Age Validation Error:



Screenshot 4: GitHub commits



Screenshot 5:Github Branches



Github Evidence:

The project used GitHub for collaboration.

Each member contributed using separate branches:

- UI Developer

- Logic Developer

- Project Manager

- Tester & Documentation

Commits and branches will be shown in screenshots.

Conclusion:

The Ticket Application System was successfully developed and tested according to the project requirements. The application correctly calculates ticket prices based on category and distance, while applying the required discount rules for age and gender.

All validation checks were implemented to ensure accurate and reliable user input, preventing invalid data from being processed. Testing confirmed that the system performs as expected, with all features functioning correctly.

GitHub was effectively used for collaboration, with each group member contributing through their assigned roles and branches. The project demonstrates a clear understanding of software development, testing, and version control practices.

Overall, the system meets all functional and non-functional requirements and is ready for submission.